By defn. of biconditional, for any arbitrary element c_2, $P(c_2)$ is True if and only if $c_1 = c_2$, for some c_1 in the domain. If for any $c_2 \neq c_1$, then $P(c_2)$ is False.

Suppose, $c_3 \neq c_1$ and $(P(c_2) \longleftrightarrow c_3 = c_2)$ for some c_3 be True.

Since c_2 is arbitrary, let $c_2 = c_1$. $\therefore$ $\boxed{P(c_1) \longleftrightarrow c_3 = c_1}$ is True.

$\because$ $c_3 \neq c_1$, by defn. of biconditional, $P(c_1)$ is False.

$\therefore$ c_1 is the only unique element for which $\exists x \forall y (P(y) \longleftrightarrow x = y)$ is True. □

b) $\exists x P(x) \land \forall x \forall y (P(x) \land P(y) \longrightarrow x = y)$

Let, $\exists x P(x) \land \forall x \forall y (P(x) \land P(y) \longrightarrow x = y)$ be True.

Solution: Let, $\exists x P(x) \land \forall x \forall y (P(x) \land P(y) \longrightarrow x = y)$ is True.

By defn. of AND, $\exists x P(x)$ is True and $\forall x \forall y (P(x) \land P(y) \longrightarrow x = y)$ is True.

- By defn. of existential quantifier, there exists some element c_1 in the domain such that $P(c_1)$ is True.
- By defn. of universal quantifier, for any arbitrary element c_2 in the domain, $\forall y (P(c_2) \land P(y) \longrightarrow c_2 = y)$ is True.
- By defn. of universal quantifier, for any arbitrary element c_3 in the domain $(P(c_2) \land P(c_3) \longrightarrow c_2 = c_3)$ is True.
- Let c be some other element compared to c_1 $(c \neq c_1)$ s.t. $P(c)$ is True.
- By defn. of AND, $P(c_1) \land P(c)$ is True.
- By Modus ponens, $(c_1 = c)$. $\therefore$ Our assumption that $c \neq c_1$ is False.

$\therefore$ c_1 is the only unique element for which $\exists x P(x) \land \forall x \forall y (P(x) \land P(y) \longrightarrow x = y)$ is True. □

c) $\exists x (P(x) \land \forall y (P(y) \longrightarrow x = y))$

Let, $\exists x (P(x) \land \forall y (P(y) \longrightarrow x = y))$ be True.

Solution: Let, $\exists x (P(x) \land \forall y (P(y) \longrightarrow x = y))$ be True.

By defn. of existential quantifier, there exists some element c_1